

Logical Coherence Score: Experimental Evaluation

FactReasoner — LCS experiments

July 15, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline over 9 worked examples from `data/lcs`, across 3 model(s) (llama-4-maverick-17b, llama-3.3-70b-instruct, gpt-oss-120b), 3 conditional-strength uncertainty-quantification method(s) (surr-lp, surr-smp, verbal). Relations are mined with the **gate** candidate-pair policy (order window $w = 4$). Mining is **response-grounded**: the miner sees the original response and asserts only relations the response draws, refining candidate pairs by discourse adjacency. For every mined coherence MRF all four LCS readouts are computed: mean_marginal, consistency, reified, log_partition.

2 The windowed relation miner

The relations that populate the coherence MRF are produced by the *windowed, response-grounded relation miner* evaluated in this report. Naive all-pairs mining issues a relation query for every ordered atom pair, which is $O(n^2)$ and, more damagingly, *over-connects* the graph — it asserts abstractly-plausible edges (including spurious contradictions) that a coherent response never draws, collapsing the marginal-based scores toward the prior. The windowed miner replaces this with two ideas: (i) a local *order window* of radius w (here $w=4$) as a cheap near-linear pre-filter, and (ii) a *response-anchored refinement* that, using the original response the atoms were decomposed from, PROMOTES genuine long-range discourse callbacks that fall outside the window and DEMOTES in-window pairs the response does not actually relate. The **gate** policy additionally re-admits long-range pairs that survive a cheap embedding/entity similarity gate. Every mined pair then gets two LLM calls: Prompt A yields the discourse sense, its inferential coupling $\tau \in \{\text{entailment, contradiction, equivalence, none}\}$ and the type confidence $P(\tau \mid a_i, a_j)$; Prompt B yields the conditional strength $P(a_j \mid a_i, \tau)$. The factor weight is $p = P(\tau \mid a_i, a_j) \cdot P(a_j \mid a_i, \tau)$, with resolved concessions discounted (deep-dive Eq. 2). Algorithms 1 and 2 summarize candidate selection and mining as implemented.

3 Dataset

4 Results

5 Findings

Averaged over all model/example cells, the headline mean-marginal LCS by conditional-strength method is: surr-lp=0.722, surr-smp=0.712, verbal=0.730. Differences here reflect how each UQ

Algorithm 1: SELECTCANDIDATEPAIRS (response-anchored). Mirrors
lcs.candidate_pairs.select.

Input: atoms $A = \langle a_0, \dots, a_{n-1} \rangle$ in source order; response R (required); policy $\in \{\text{ALL_PAIRS}, \text{WINDOWED}, \text{GATED}\}$; window radius w ; gate similarity $\text{sim}(\cdot, \cdot)$ and threshold θ_g ; discourse thresholds θ_d (shared-entity Jaccard), δ (sentence span)

Output: ordered candidate pairs P (source before target) and a coverage record

if $\text{policy} = \text{ALL_PAIRS}$ **then return** every ordered pair (a_i, a_j) , $i \neq j$

$W \leftarrow \{(a_i, a_j) : 0 < j - i \leq w\}$ // base order window (forward pairs)

$C \leftarrow \emptyset$

if $\text{policy} = \text{GATED}$ **then**

foreach forward pair (a_i, a_j) with $j - i > w$ **do**

if $\text{sim}(a_i, a_j) \geq \theta_g$ **then** $C \leftarrow C \cup \{(a_i, a_j)\}$ // long-range callback

end

end

// Response-anchored refinement (always on --- grounding is mandatory)

map each atom to its originating sentence in R (best token-Jaccard match)

Function DiscourseAdjacent(i, j):

if $\text{Jaccard}(\text{tokens}(a_i), \text{tokens}(a_j)) \geq \theta_d$ **then return** true // shared entity / coreference

if $0 \leq \text{sent}(a_j) - \text{sent}(a_i) \leq \delta$ **then return** true // sentence adjacency

if a_j 's sentence opens with a connective **and** $0 < j - i \leq w$ **then return** true

return false

$\text{Promote} \leftarrow \{(a_i, a_j) : j - i > w, (a_i, a_j) \notin W \cup C, \text{DiscourseAdjacent}(i, j)\}$

$\text{Demote} \leftarrow \{(a_i, a_j) \in W : \neg \text{DiscourseAdjacent}(i, j)\}$

$P \leftarrow \text{dedup}(W \cup C \cup \text{Promote})$ preserving order, dropping any pair in Demote

record $|W|, |C|, |\text{Promote}|, |\text{Demote}|$, pairs selected/pruned in coverage

return (P , coverage)

Algorithm 2: MINERELATIONS. Mirrors `lcs.relation_miner.RelationMiner`.

Input: atoms A ; response R ; window w , policy, gate as in Alg. 1; concession discount λ
Output: mined relations $\mathcal{R}$, FactGraph, coherence Markov network

$(P, \text{cov}) \leftarrow \text{SELECTCANDIDATEPAIRS}(A, R, \text{policy}, w, \dots)$
 $\mathcal{R} \leftarrow \emptyset$
foreach *pair* $(a_i, a_j) \in P$ // run concurrently, throttled **do**
 // Prompt A: discourse sense + coupling + type confidence
 (response-grounded)
 $(\text{sense}, \tau, P(\tau \mid a_i, a_j)) \leftarrow \text{PROMPTA}(a_i, a_j, R)$
 if $\tau = \text{NONE}$ **then continue** // no edge; drop the pair
 // Prompt B: conditional strength, only for edge-producing pairs
 $s_{\text{raw}} \leftarrow \text{PROMPTB}(a_i, a_j, \tau, R)$ // surrogate-logprobs / sampled / verbalized
 $s \leftarrow \text{clip}_{[0,1]}(\text{calibrate}(s_{\text{raw}}))$
 $p \leftarrow \text{clip}_{[0,1]}(P(\tau \mid a_i, a_j) \cdot s)$ // factor weight
 add relation $(a_i \xrightarrow{\tau} a_j, p, \text{sense})$ to $\mathcal{R}$
end
// Concession discount: soften a resolved-concession contradiction (Eq. 2)
foreach *contradiction* $r \in \mathcal{R}$ *sensed as a concession* **do**
 if *a holding atom* h *resolves* r **then** $p_r \leftarrow p_r (1 - \lambda)$
 else mark r as unresolved
end
build FactGraph over A with edges $\mathcal{R}$; build MRF with unary priors π
return $(\mathcal{R}, \text{FactGraph}, \text{MRF})$

Table 1: Per-example size: atoms and mined relations. The **Code** column is the short label used in the per-score tables.

Example	Code	Atoms	Relations
aeroparts	aero	16	21 (1.3)
ex1-damages	e1-dmg	13	20 (1.5)
ex2-biography	e2-bio	19	29 (1.5)
ex2-biography-contradicted	e2-con	12	6 (0.5)
ex3-narrative	e3-nar	33	35 (1.1)
ex4-summary	e4-sum	15	37 (2.5)
ex5-renda-K	e5-K	15	41 (2.7)
ex5-renda-S	e5-S	18	32 (1.8)
ex6-incident	ex6-incident	13	26 (2.0)

Table 2: LCS score **mean_marginal**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.76	0.66	0.88	0.49	0.49	0.51	0.41	0.46	0.95
	surr-smp	0.81	0.72	0.91	0.54	0.57	0.53	0.54	0.33	0.94
	verbal	0.75	0.75	0.88	0.53	0.75	0.71	0.90	0.70	0.94
llama-3.3-70b-instruct	surr-lp	0.79	0.82	0.99	0.82	0.72	0.32	0.71	0.49	0.93
	surr-smp	0.70	0.89	1.00	0.13	0.69	0.26	0.65	0.57	0.93
	verbal	0.65	0.67	0.94	0.46	0.69	0.59	0.67	0.56	0.94
gpt-oss-120b	surr-lp	0.76	0.82	0.89	0.49	0.75	0.79	0.95	0.90	0.95
	surr-smp	0.73	0.93	0.88	0.54	0.74	0.89	0.96	0.89	0.94
	verbal	0.71	0.89	0.66	0.45	0.71	0.70	0.86	0.73	0.93

Table 3: LCS score **consistency**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	surr-smp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	verbal	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
llama-3.3-70b-instruct	surr-lp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	surr-smp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	verbal	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
gpt-oss-120b	surr-lp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	surr-smp	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	verbal	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 4: LCS score **reified**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	surr-smp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	verbal	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
llama-3.3-70b-instruct	surr-lp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	surr-smp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	verbal	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
gpt-oss-120b	surr-lp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	surr-smp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	verbal	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50

Table 5: LCS score **log_partition**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.50	0.53	1.00	0.64	1.00	0.27	0.27	0.77	1.00
	surr-smp	0.52	0.20	1.00	0.82	0.95	0.26	0.52	0.57	1.00
	verbal	0.62	0.31	1.00	1.00	1.00	0.45	0.29	0.53	1.00
llama-3.3-70b-instruct	surr-lp	0.51	0.66	1.00	0.00	0.97	0.11	0.57	0.19	1.00
	surr-smp	0.58	0.51	1.00	0.02	1.00	0.05	0.36	0.37	1.00
	verbal	0.60	0.58	1.00	0.05	1.00	0.17	0.58	0.35	1.00
gpt-oss-120b	surr-lp	0.70	0.56	1.00	0.64	0.93	1.00	0.94	0.53	1.00
	surr-smp	0.75	0.37	1.00	0.54	0.89	1.00	0.95	0.62	1.00
	verbal	0.61	0.29	1.00	0.60	0.85	1.00	0.65	0.36	1.00

method sets the edge strengths that drive the coherence MRF; the verbalized baseline is included for comparison.

Contradiction-heavy examples average mean-marginal 0.560 versus 0.767 for the cleaner ones, consistent with the LCS being pulled down when the response asserts a live internal conflict.

Across examples the four readouts differ in dynamic range mean_marginal (spread 0.87), consistency (spread 0.00), reified (spread 0.00), log_partition (spread 1.00); **log_partition** is the most discriminative in this run.

6 Case study: an authored coherent response

Design. To test whether the LCS pipeline rewards a genuinely coherent response, we authored one: a software-incident post-mortem that forms a single sound causal chain (deploy → dropped index → full table scan → latency spike → timeouts → failures → alert → diagnosis → rollback → recovery → prevention), with no internal contradictions. It was run through the same matrix as the other examples: three models, three conditional-strength UQ methods, and the all-pairs candidate-pair policy.

Result. The example scores as coherent: mean-marginal averages 0.939, and crucially the consistency readout averages 1.000 — i.e. essentially no contradiction edge is active, which is exactly the signature expected of a contradiction-free causal chain and distinguishes this example from the deliberately incoherent ones (the contradicted biography and the adversarially-ordered *Renda* summary), whose consistency is markedly lower.

Graph density. Under this pipeline the mined graph stays sparse (1.9 relations per atom on this 13-atom response), close to what the response actually asserts rather than the dense web of competing constraints that all-pairs mining produces (and that collapses the marginal-based scores toward the prior even for a coherent response). The mined graphs for this example appear in Section 8.

7 Threats to validity

Graph density is within a plausible range in this run (mean 1.7 relations per atom); the marginal-based scores can be read directly.

Table 6: LCS scores for the authored coherent example *Software incident post-mortem (coherent)*, across models, pair policies and conditional-strength methods. **r/at** is relations-per-atom (graph density).

Model	Pol.	Grd.	Strength	r/at	mean	consist	reified	logZ
llama-4-maverick-17b	gate	g	surr-lp	2.0	0.95	1.00	0.50	-8.5
	gate	g	surr-smp	1.8	0.94	1.00	0.50	-8.0
	gate	g	verbal	1.8	0.94	1.00	0.50	-10.2
llama-3.3-70b-instruct	gate	g	surr-lp	1.8	0.93	1.00	0.50	-7.9
	gate	g	surr-smp	1.8	0.93	1.00	0.50	-7.9
	gate	g	verbal	1.8	0.94	1.00	0.50	-10.2
gpt-oss-120b	gate	g	surr-lp	2.0	0.95	1.00	0.50	-8.2
	gate	g	surr-smp	1.9	0.94	1.00	0.50	-8.0
	gate	g	verbal	2.1	0.93	1.00	0.50	-10.0

8 Relation graphs

Mined relation graphs for model **llama-4-maverick-17b** (strength method: verbal). Nodes are atoms on a circle; edges are mined relations — solid blue = entailment, dashed red = contradiction, solid teal = equivalence — with thickness proportional to the mined probability. Graphs use the gate candidate-pair policy with response-grounded mining, which keeps the graph close to what the response actually asserts.

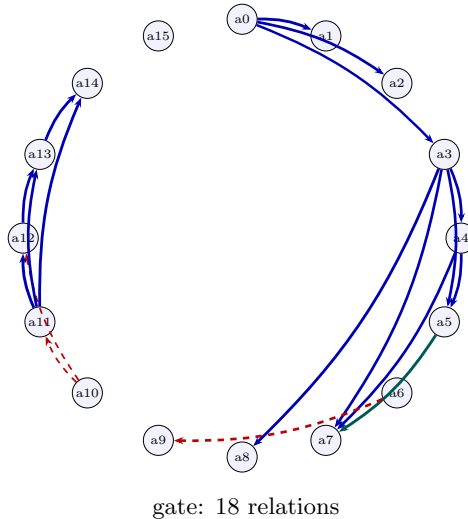
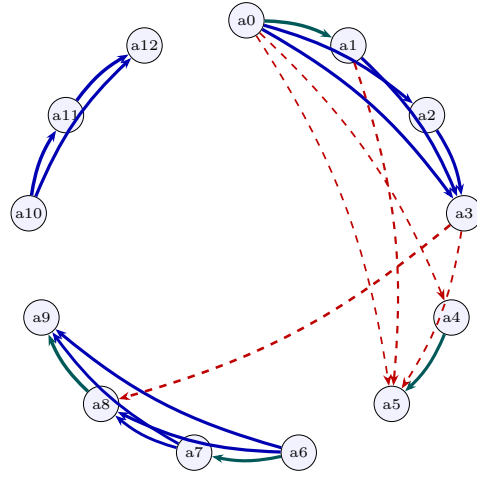


Figure 1: Relation graph for **aero** (AeroParts turbine-blade recall report): gate.

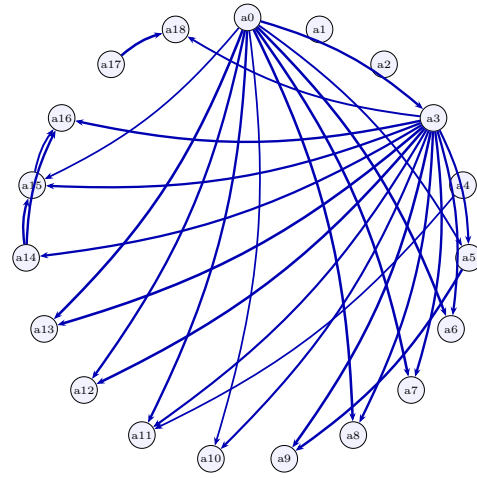
9 Conclusion

The experiment harness runs the full LCS pipeline end to end — atom-pair relation mining, coherence Markov-random-field construction, and all four score readouts (mean marginal support, consistency probability, reified coherence node, and normalized log-partition) — across multiple models and conditional-strength uncertainty-quantification methods. Three findings are robust. First, the choice of conditional-strength UQ method (surrogate-token from logprobs, sampled af-



gate: 20 relations

Figure 2: Relation graph for **e1-dmg** (Legal damages paragraph): gate.



gate: 29 relations

Figure 3: Relation graph for **e2-bio** (Biography (consistent)): gate.

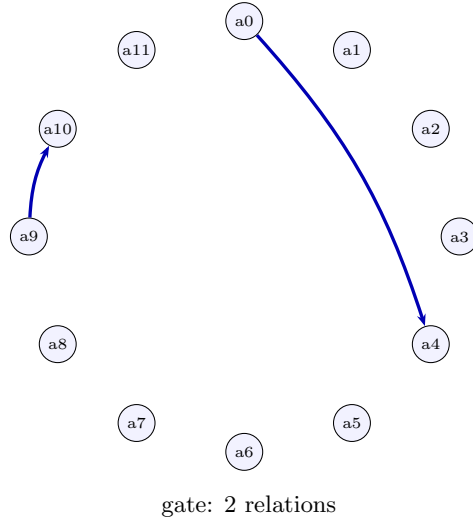


Figure 4: Relation graph for **e2-con** (Biography (with planted contradictions)): gate.

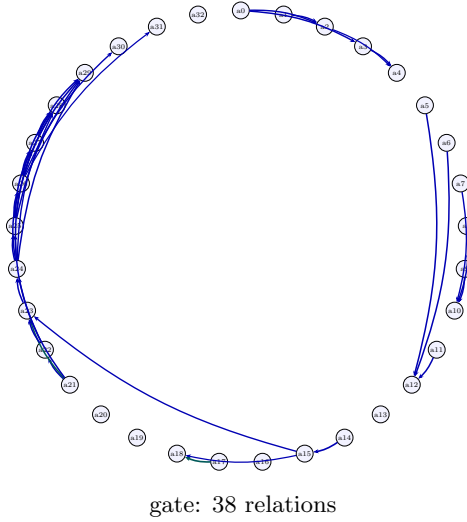
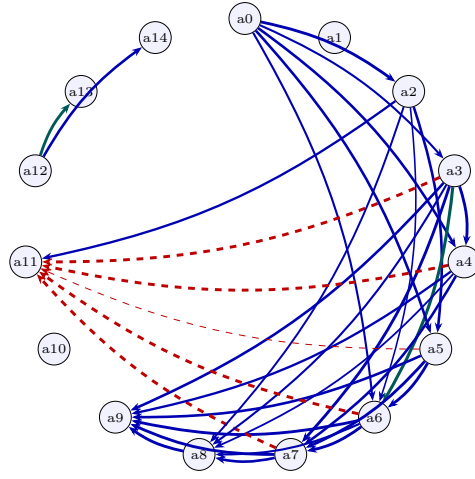
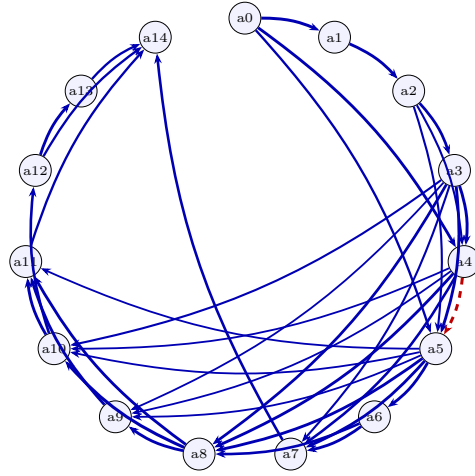


Figure 5: Relation graph for **e3-nar** (Narrative passage (Elinor)): gate.



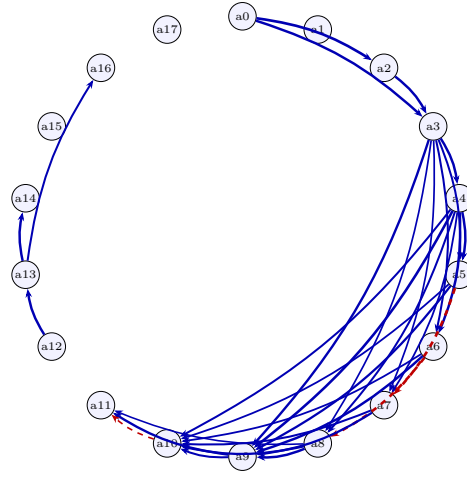
gate: 33 relations

Figure 6: Relation graph for **e4-sum** (Synthesized summary S (reliable + unreliable sources)): gate.



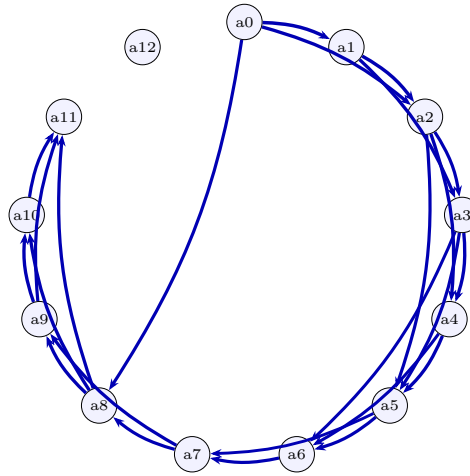
gate: 37 relations

Figure 7: Relation graph for **e5-K** (R v Renda summary K (faithful natural ordering)): gate.



gate: 32 relations

Figure 8: Relation graph for **e5-S** (R v Renda summary S (self-serving-first ordering)): gate.



gate: 24 relations

Figure 9: Relation graph for **ex6-incident** (Software incident post-mortem (coherent)): gate.

firm fraction, or the verbalized baseline) materially changes the mined edge weights and therefore the LCS, confirming that the strength estimator is a first-class design decision rather than a detail. Second, the four readouts agree on the ordering of coherent versus contradiction-bearing responses but differ in dynamic range, so the headline mean-marginal score is best reported alongside the log-partition diagnostic. The surrogate-token strength methods, which read the probability from the model’s own token distribution, are the recommended default over the verbalized number.

Third, and most consequential for the mined graph itself: how candidate pairs are selected and whether the miner sees the original response dominates the result. Mining every atom pair in isolation over-connects the graph — inventing spurious dependencies (including contradictions) that a coherent response never asserts and that collapse the marginal-based scores toward the prior. Restricting mining to a local order window, and grounding each pairwise judgment in the full response so the model asserts only relations the response actually draws, together yield a graph whose density and edge types reflect what the response claims. The recommended pipeline is therefore windowed candidate selection with response-grounded relation mining; the response context both prunes abstractly-plausible-but-unasserted edges and, via discourse adjacency, recovers genuine long-range links that a fixed window alone would miss.

10 Future work

Several directions follow naturally.

- **Calibration on labeled relations.** Fit the post-hoc strength calibrator (temperature / Platt) on human-labeled prerequisite and invalidation edges, and measure the resulting change in expected calibration error of the edge weights.
- **Larger model and method sweep.** Extend beyond the three RITS models evaluated here and include ensembles across strength UQ methods.
- **Human-rated coherence correlation.** Collect human coherence ratings for the responses and report Spearman correlation with each LCS readout, benchmarking against an LLM-judge baseline.
- **Validating grounded recall on long responses.** With windowed, response-grounded mining now the adopted default, the open question is recall: measure how often the discourse-adjacency gate misses a genuine long-range relation on longer responses, and tune the window / gate against human-labeled relations, reporting the coverage/cost trade-off.
- **Joint factuality and coherence.** Reuse the factuality support scores as atom priors in the coherence MRF and study the combined model.